

Industrial Internship Report on ServiceHub – Multi-Tenant Service Marketplace

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on developing a practical solution for the assigned project within the six-week internship period. The project involved understanding requirements, designing the application, implementing the required features, testing the system and documenting the overall work.

My project was **ServiceHub – Multi-Tenant Service Marketplace**, a web-based platform designed to connect customers with service providers through a centralized service marketplace. The system supports Customer, Merchant and Admin roles, allowing customers to discover and book services, merchants to manage their services and bookings, and administrators to manage users, services and bookings.

The project provided me with practical exposure to full-stack web development using React, Node.js, Express.js and MongoDB. It also helped me understand REST APIs, authentication, role-based access control, database management, service booking workflows, testing and GitHub-based project submission.

This internship was a valuable opportunity to gain practical experience in software development and apply technical concepts to a real-world project. It improved my problem-solving, development and project-management skills and provided an overall meaningful learning experience.

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1 Preface

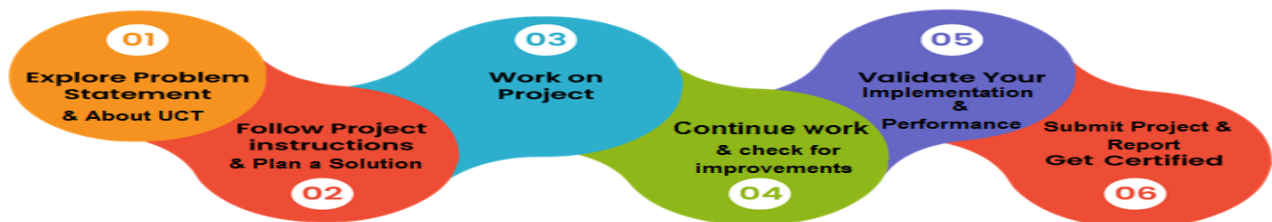
The six-week Industrial Internship provided me with an opportunity to understand, develop and test a practical full-stack web application. The internship involved understanding the problem statement, planning the solution, implementing the application, improving the system, validating the implementation and preparing the final project and report.

An industry-oriented internship is important for career development because it provides practical exposure beyond academic learning and helps students apply technical concepts to real-world problems.

My project, ServiceHub – Multi-Tenant Service Marketplace, is a web-based platform that connects customers with service providers. It supports Customer, Merchant and Admin roles and provides service discovery, service management, booking, authentication and administration features.

The opportunity provided by upskill Campus, The IoT Academy and UniConverge Technologies Pvt Ltd (UCT) helped me gain practical exposure to software development and understand how a project is completed within a defined timeline.

The program was planned in stages, beginning with understanding the problem and project requirements, followed by implementation, continuous improvement, validation and final project and report submission.



The internship strengthened my understanding of full-stack development and provided practical experience with React, Node.js, Express.js and MongoDB. I also learned about REST APIs, authentication, role-based access control, database operations, testing and GitHub-based project submission.

I sincerely thank upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd (UCT), my college Sai Vidya Institute of Technology, and everyone who supported and encouraged me throughout the internship.

I encourage my juniors and peers to make the most of internship opportunities by actively learning, practicing technical concepts and completing projects with dedication. Practical experience and consistent effort can greatly contribute to career development.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT)**, **Cyber Security**, **Cloud computing (AWS, Azure)**, **Machine Learning**, **Communication Technologies (4G/5G/LoRaWAN)**, **Java Full Stack**, **Python**, **Front end** etc.

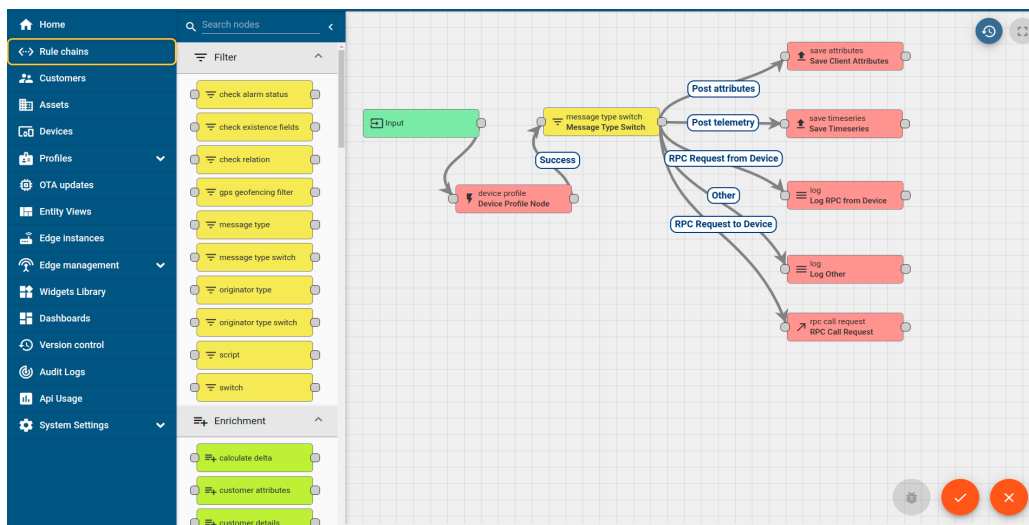
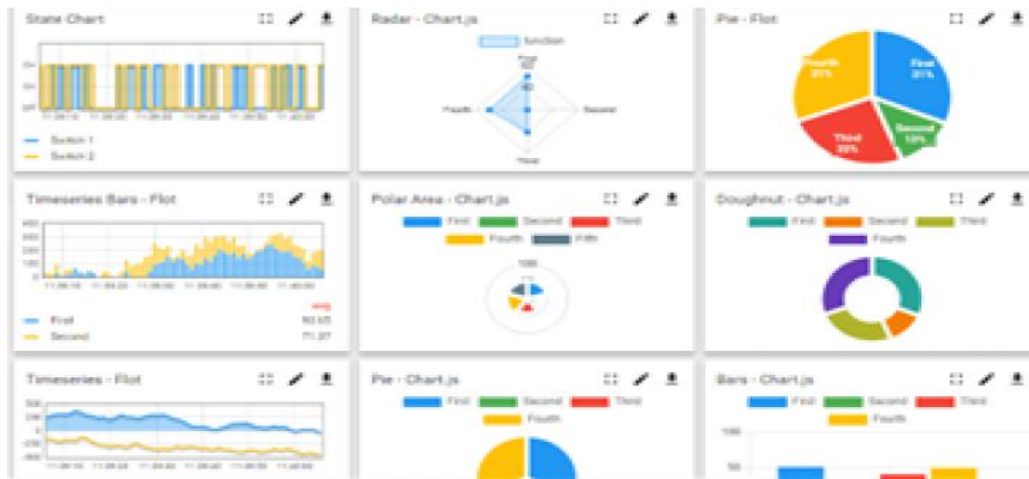


i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.
- It has features to
 - Build Your own dashboard
 - Analytics and Reporting
 - Alert and Notification
 - Integration with third party application (Power BI, SAP, ERP)
 - Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



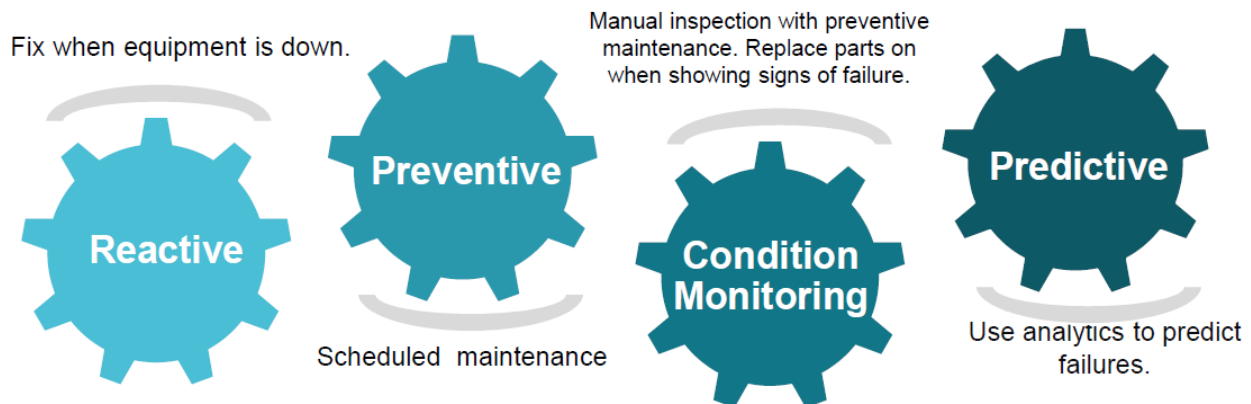


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

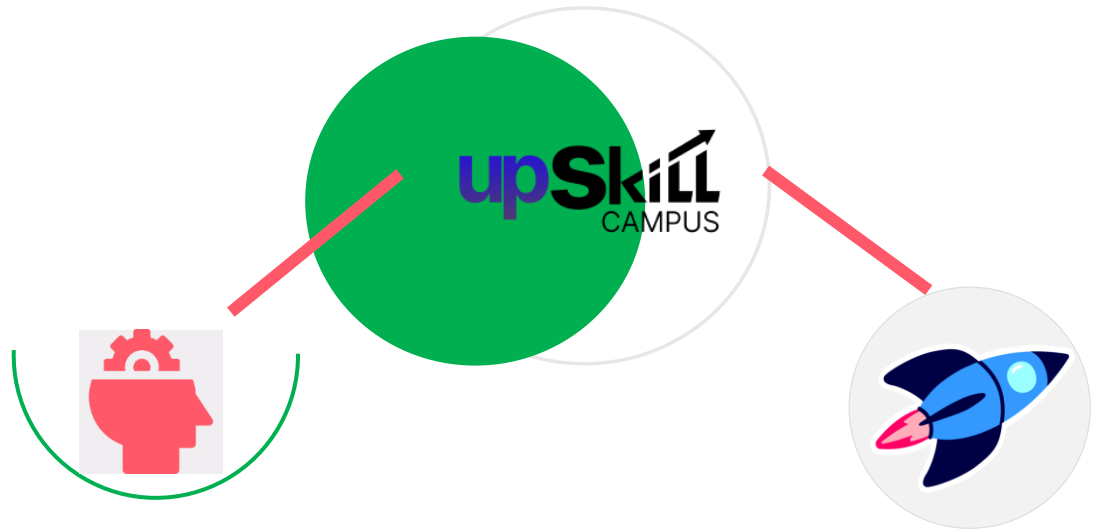
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

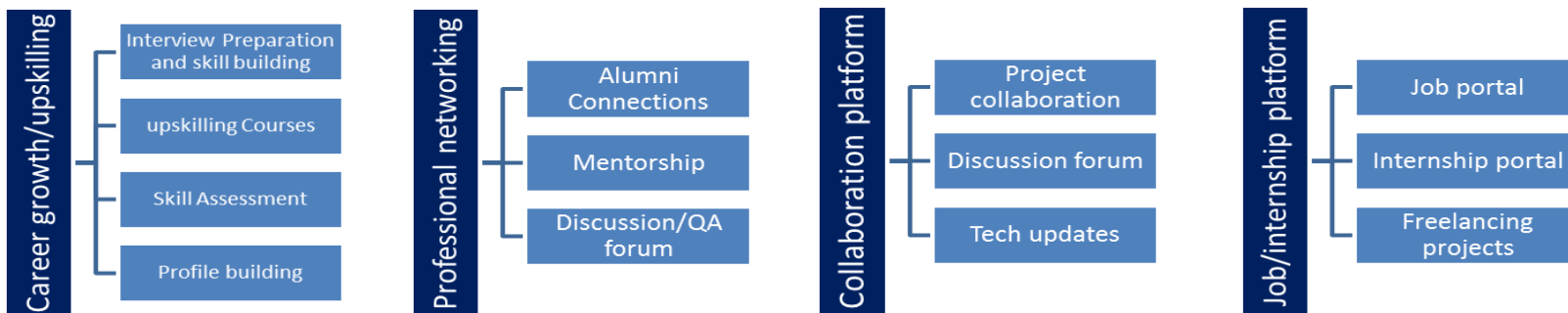
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objectives of this internship program were to:

- Gain practical experience in full-stack web application development.
- Understand and solve a real-world service marketplace problem.
- Develop a multi-tenant web application with Customer, Merchant and Admin roles.
- Gain practical experience with React, Node.js, Express.js and MongoDB.
- Understand REST APIs, authentication, role-based access control and database operations.
- Develop practical skills in service discovery, booking workflows and service management.
- Improve problem-solving, testing, documentation and project submission skills.
- Gain industry exposure and improve career readiness through practical project development.

2.5 Reference

[1] React Documentation – React official documentation for building user interfaces and frontend applications.

[2] Node.js and Express.js Documentation – Official documentation for Node.js runtime and Express.js backend development and REST APIs.

[3] MongoDB and Mongoose Documentation – Official documentation for MongoDB database management and Mongoose object data modeling.

2.6 Glossary

Terms	Acronym
Application Programming Interface	API
Representational State Transfer	REST
JSON Web Token	JWT
Create, Read, Update and Delete	CRUD
Database	DB

3 Problem Statement

The assigned problem was to develop ServiceHub – Multi-Tenant Service Marketplace, a web-based platform that provides a centralized way for customers to discover and book services from service providers.

In a traditional service booking process, customers may need to search for service providers through different sources, compare available services and contact providers separately to arrange appointments. Service providers also need an organized way to manage their services and customer bookings.

ServiceHub addresses this problem by providing a single platform for service discovery, service management and booking. The system supports three roles: Customer, Merchant and Admin. Customers can explore service categories, search for services and make bookings. Merchants can manage their services and view their bookings, while administrators can manage users, services and bookings.

The application was developed as a full-stack web solution using React for the frontend, Node.js and Express.js for the backend, and MongoDB with Mongoose for database management. Authentication and role-based access control were implemented to provide appropriate access to different users.

The objective of the solution is to provide a structured, convenient and centralized service marketplace that improves the service discovery and booking experience for customers while helping merchants and administrators manage the platform effectively.

4 Existing and Proposed solution

Existing Solution

Customers commonly find service providers through separate websites, social media, phone calls or local contacts. Service providers may also use different methods to manage their services and customer bookings. These approaches can make service discovery, comparison and booking less organized and convenient.

Proposed Solution

ServiceHub – Multi-Tenant Service Marketplace provides a centralized web-based platform for discovering and booking services. The system supports three roles: Customer, Merchant and Admin. Customers can browse categories, search for services and make bookings. Merchants can manage their services and view customer bookings, while Admins can manage users, services and bookings.

Value Addition

The proposed solution provides a structured service marketplace with centralized service discovery and booking. It provides role-based access, authentication, service management, booking management and database integration. This improves convenience for customers and provides merchants and administrators with an organized platform for managing services and bookings.

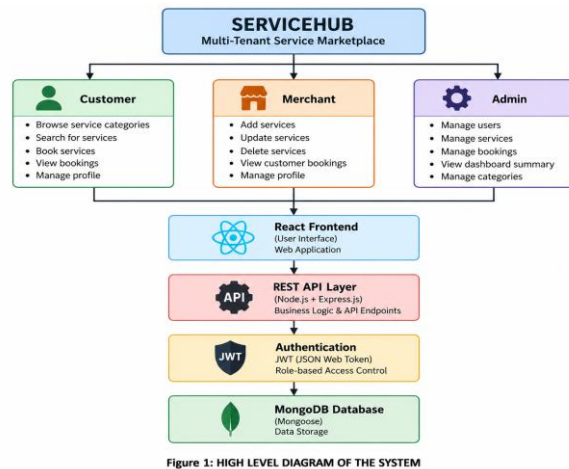
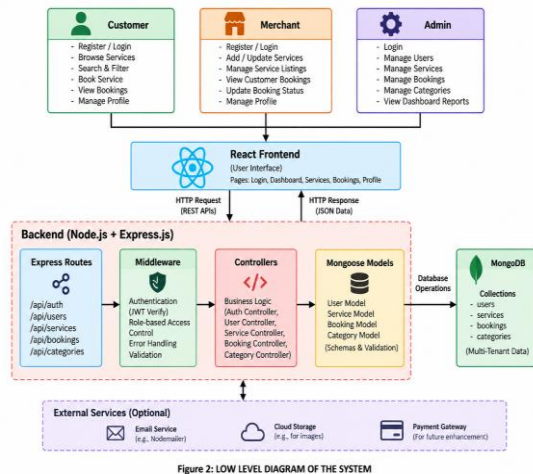
4.1 Code submission (Github link)

<https://github.com/sravanthiks24is-oss/upskillcampus>

5 Proposed Design/ Model

5.1 High Level Diagram

5.2 Low Level Diagram



5.3 Interfaces

The ServiceHub interfaces were designed to provide simple and role-specific interaction with the system.

Customer Interface:

Customers can register and log in, browse service categories, search for services, view service details, select appointment dates and times, provide booking details and view their bookings.

Merchant Interface:

Merchants can log in to their dashboard, add new services, update service information, delete their own services and view customer bookings related to their services.

Admin Interface:

Administrators can access the admin dashboard to view system summaries and manage users, services and bookings.

The frontend interfaces communicate with the backend through REST APIs. Authentication and role-based access control are used to provide appropriate access to Customer, Merchant and Admin users.

6 Performance Test

The performance testing of ServiceHub focused on validating the reliability, usability and correctness of the implemented web application features. The main constraints considered during development were authentication and authorization, service availability, booking validation, database operations, role-based access control and frontend-backend communication. These constraints were addressed through JWT-based authentication, role-based middleware, input validation, future date and time validation, active service checks, MongoDB database operations and REST API integration. The implemented system was tested through functional test cases covering Customer, Merchant and Admin workflows. The testing included user authentication, service discovery, service management, booking operations, GPS location capture and dashboard access. The testing results showed that the major implemented workflows operated as expected under the tested conditions. Identified issues during development were corrected through debugging and repeated testing.

6.1 Test Plan/ Test Cases

Test Case	Feature Tested	Expected Result	Actual Result	Status
TC01	Customer Registration	Customer account should be created successfully	Account created successfully	Pass
TC02	Customer Login	Valid customers should allow login	Customer logged in successfully	Pass
TC03	Service Discovery	Customer should be able to browse service categories and services	Services displayed correctly	Pass
TC04	Service Search	Search should return the relevant service	Service search worked correctly	Pass
TC05	Booking Validation	Past date/time should not be accepted	Invalid booking time was rejected	Pass
TC06	Service Booking	Customer should be able to book an available service	Booking created successfully	Pass
TC07	GPS Location	Booking location should be captured when GPS is enabled	Latitude and longitude were saved	Pass

TC08	Merchant Dashboard	Merchant should access and manage their services	Merchant operations worked correctly	Pass
TC09	Admin Dashboard	Admin should view users, services and bookings	Admin data displayed correctly	Pass
TC10	Role-Based Access	Users should access only permitted features	Unauthorized access was restricted	Pass

6.2 Test Procedure

The ServiceHub application was tested systematically using the local development environment. The backend server was started and connected to the MongoDB database. The frontend application was started and connected to the backend REST APIs. Customer registration and login were tested using valid and invalid credentials. Service discovery and search functionality were tested across different service categories. Booking functionality was tested by selecting a service, entering a future date and time, providing address and phone details, and submitting the booking. GPS location capture was tested during the booking process. Merchant functionality was tested for adding, updating and deleting services and viewing customer bookings. Admin functionality was tested for viewing users, services, bookings and dashboard summary information. Role-based access restrictions were tested to verify that users could access only the features permitted for their roles. The results were reviewed and issues identified during testing were corrected through debugging and repeated testing.

6.3 Performance Outcome

The performance testing confirmed that the major ServiceHub workflows operated correctly under the tested conditions. Customer registration and login worked successfully with authentication and role-based access control. Customers were able to browse categories, search for services, view service details and create bookings using valid future dates and times. Booking validation successfully prevented invalid past date and time selections. Booking details including address, phone number and GPS coordinates were stored correctly when location access was available. Merchant operations including service creation, service update and service deletion were tested successfully. Merchant access was restricted to the merchant's permitted operations. The Admin Dashboard successfully displayed users, services and bookings. Role-based access control prevented unauthorized users from accessing protected administrative functionality.

7 My Learnings

During the six-week internship, I gained practical experience in full-stack web application development by working on ServiceHub – Multi-Tenant Service Marketplace. I learned how to develop a frontend using React and connect it with a backend built using Node.js and Express.js. I gained practical understanding of REST APIs, request handling, authentication and role-based access control. I also learned how to work with MongoDB and Mongoose for storing and managing users, services and bookings. Implementing Customer, Merchant and Admin roles helped me understand how different users can be provided with appropriate access to system features. The project improved my understanding of service discovery, search, booking workflows, service management and database operations. I also gained experience in testing, debugging and identifying issues during development. Working on the project helped me improve my problem-solving, logical thinking, documentation and project management skills. The experience will help me in my career by giving me a stronger foundation in full-stack development and practical software project development.

8 Future work scope

The following enhancements can be considered for future development of ServiceHub:

- Integrate a production-ready online payment gateway for secure service payments.
- Add service provider ratings and customer reviews to improve trust and service selection.
- Implement advanced search and filtering based on location, price, category and service availability.
- Add real-time booking notifications for customers and merchants.
- Provide improved merchant analytics and reports for service performance and bookings.
- Add stronger security features and production-level deployment configurations.
- Develop a mobile application to provide easier access to ServiceHub services.
- Improve the platform's scalability and performance to support a larger number of users and service providers.